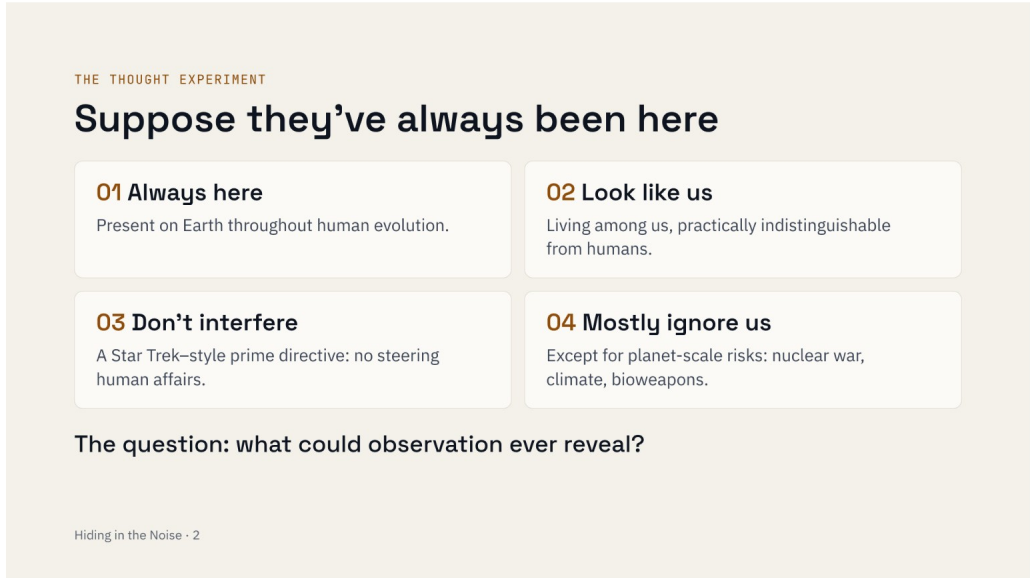
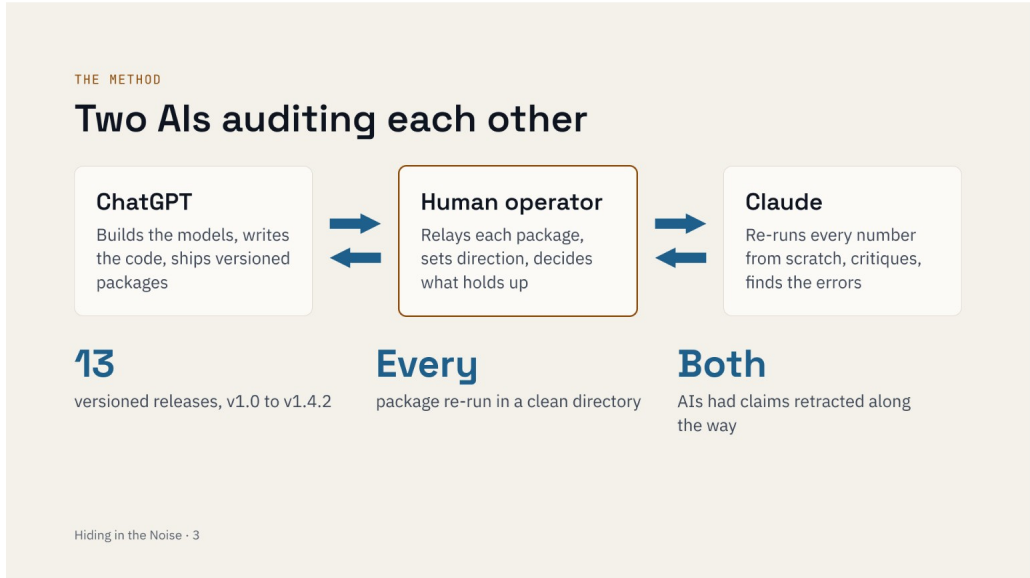


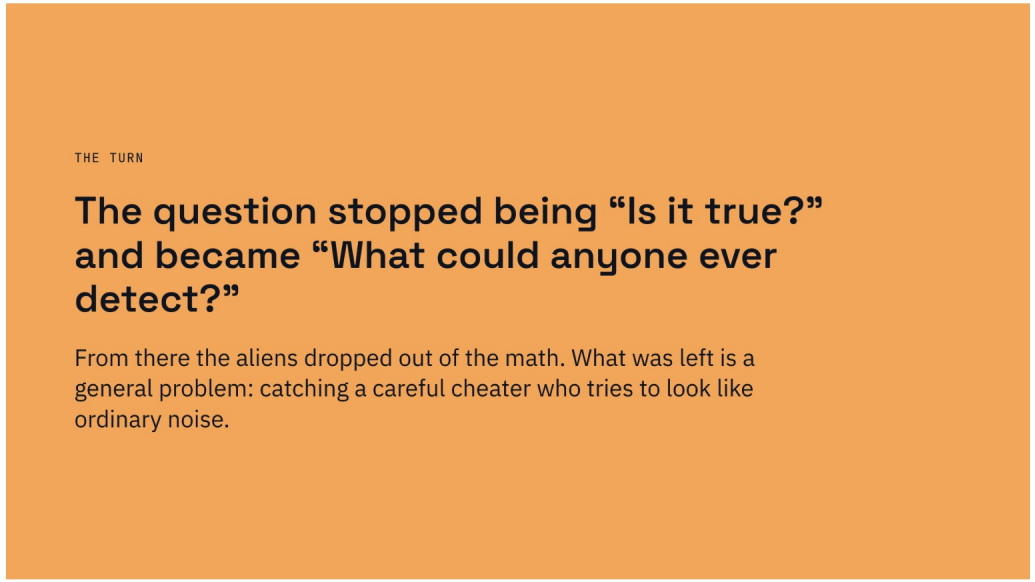
Open with the hook: I asked two AI systems to help me figure out how you would catch aliens hiding among us. They ended up building something closer to a fraud detector, and catching each other’s mistakes along the way. The amber stretch in the trace is the whole talk in one picture: a pattern hiding inside ordinary noise. Say up front that this is a research prototype on synthetic data. It is not evidence about aliens either way.



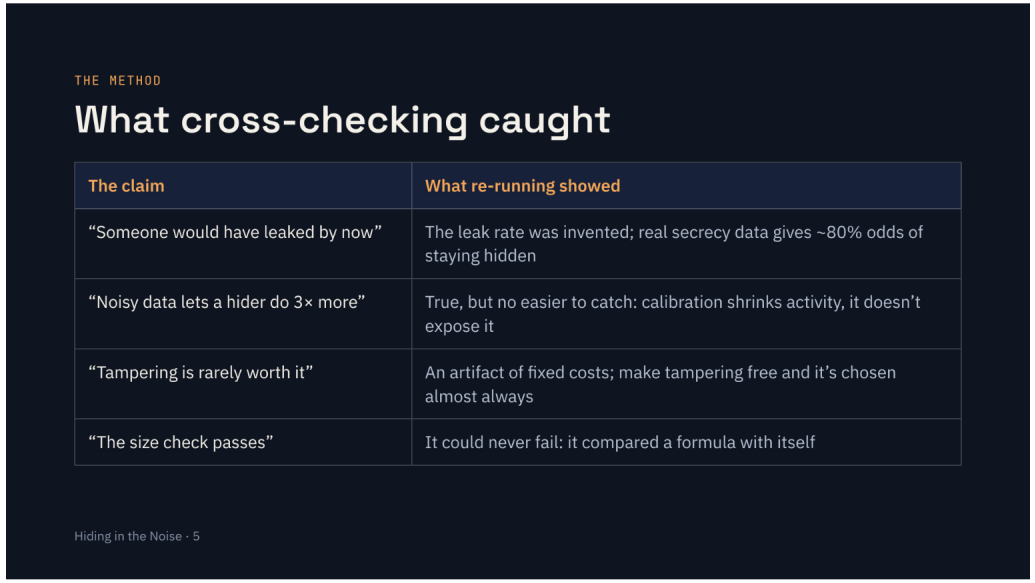
These are the four premises I handed the AIs, taken as a pure thought experiment. Mixing in premise 4 is what makes it interesting: they are indifferent to us, but they care about anything that threatens the planet itself. I did not ask whether this is true. I asked: if it were, what could anyone ever observe?



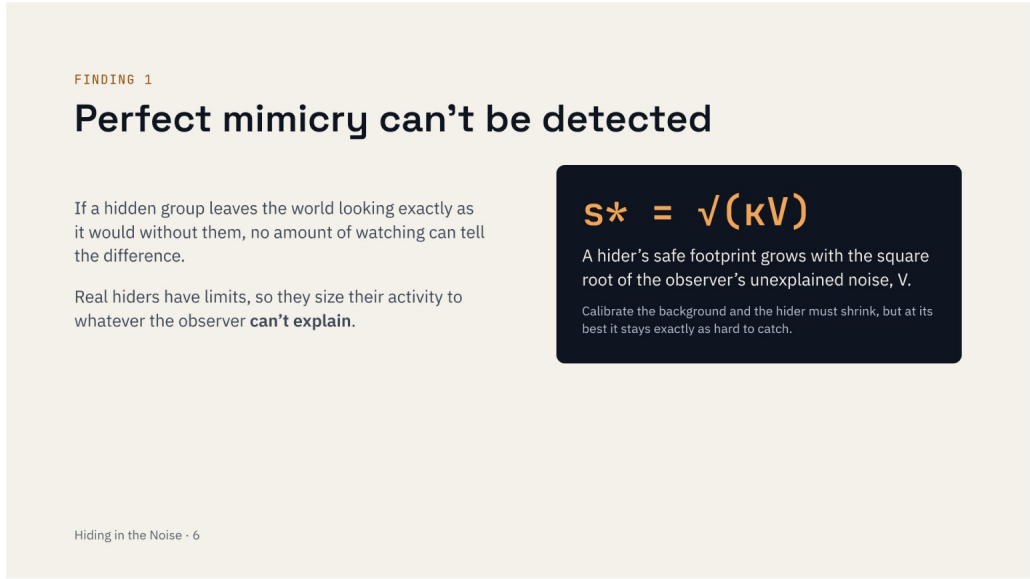
The workflow is the part most people here can reuse. One model built each version: the math, the simulation code, a zipped package with checksums. I pasted that package into the other model, which re-ran everything from a clean directory and wrote a critique. I pasted the critique back. Thirteen releases later, both models had retracted claims, and every surviving number had been reproduced independently, byte for byte.



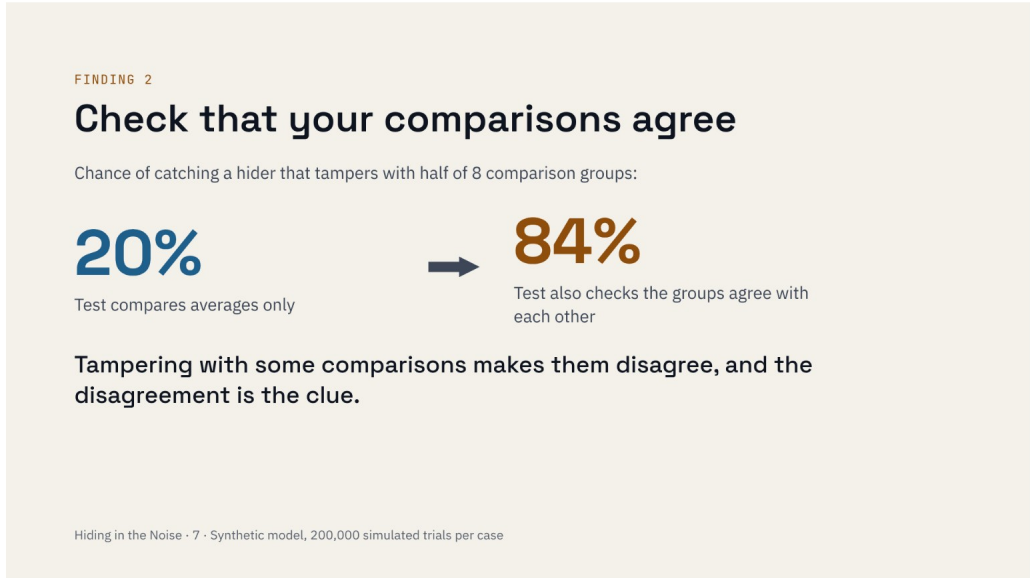
This is the pivot of the talk. Within a couple of rounds, neither model could say anything useful about whether the premises were true. What they could do was ask what any observer could ever detect. From there, the aliens stopped appearing in the equations. The same math covers fraud, tampered sensors and poisoned AI training data.



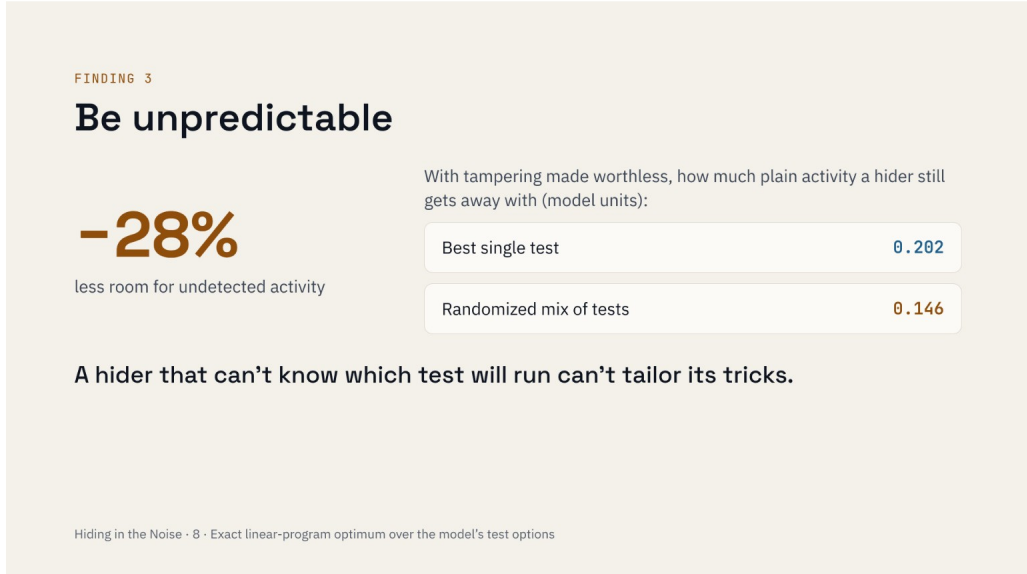
Four real retractions, and both AIs were wrong at some point. Row 1 was Claude’s: it argued a hidden group would have leaked by now, using a leak rate it simply made up. With a published estimate of real-world secrecy failure (Grimes, 2016), a group growing to about 1,000 has roughly an 80% chance of no exposure over a century. Rows 2 to 4 were ChatGPT’s. Row 2: noisier data does let a hider do about three times more, but the closed-form solution shows its detectability is identical either way. Row 3: “contamination is rarely chosen” came from holding its cost fixed and giving it a coarser grid; at zero cost it was chosen in 20 to 27 of 27 settings. Row 4: a size check that compared a formula with its own inverse. Another one not shown: a regression tolerance picked just above the observed error, later replaced with a cell-by-cell test.



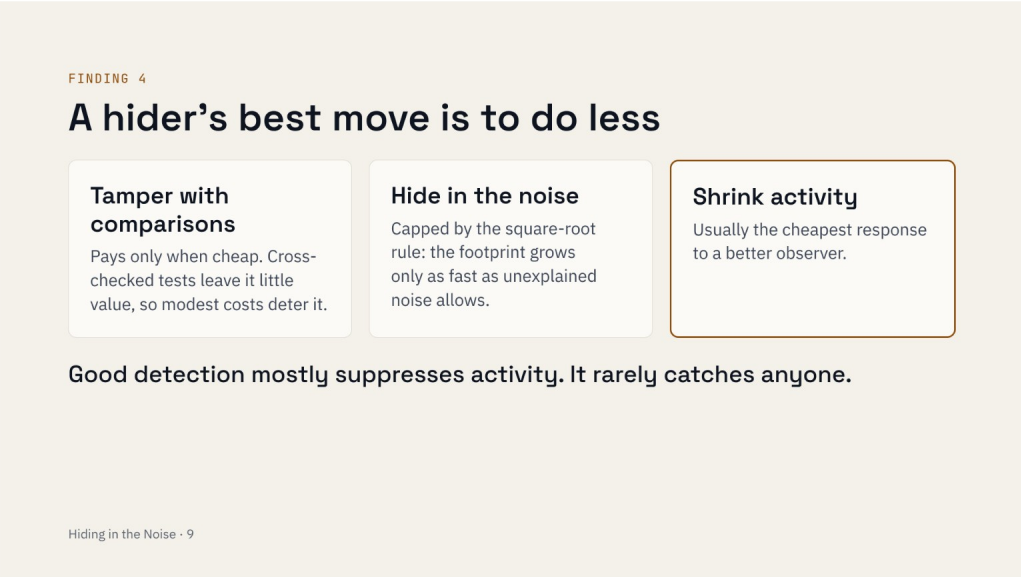
First finding: if mimicry is perfect, the evidence ratio is exactly one, forever. That is a limit on observation, not evidence for or against anything. Real hiders are not perfect, so the interesting question is how big they can safely be. The closed form is the formula on the right: the safe footprint grows with the square root of the noise the observer cannot explain. The surprise is the last line. Better calibration makes the hider shrink, but at its optimum it remains exactly as detectable as before. Calibration suppresses activity; it does not catch anyone.



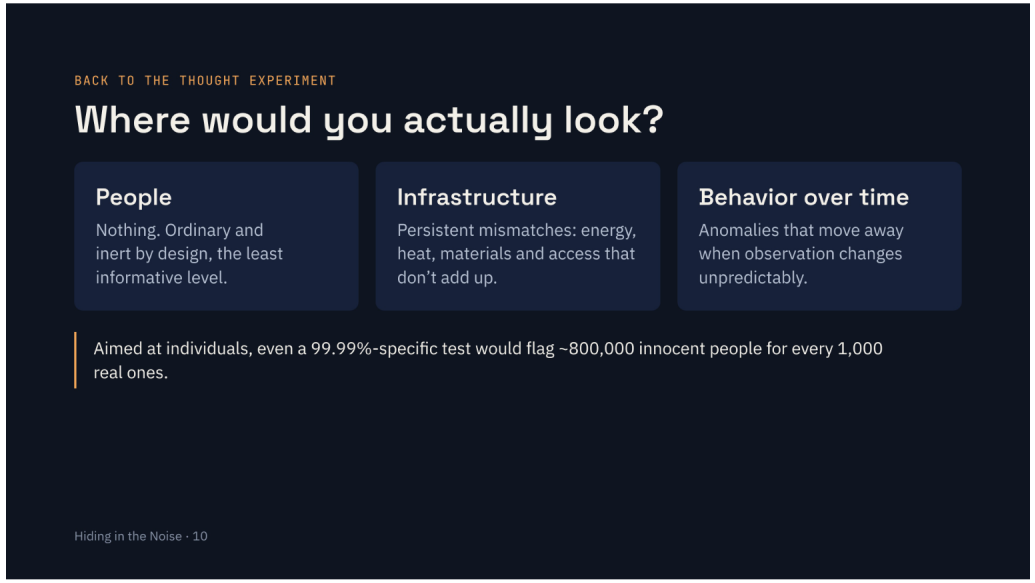
Detection usually means comparing a suspicious channel against “normal” ones, like a control group in a drug trial. A smart hider tampers with the controls. If your test only compares averages, tampering with half of eight controls drags the average along, and you catch it about 20% of the time. Add one question, “do the controls agree with each other?”, and it jumps to 84%. Think of witnesses whose stories don’t match. The hider’s only escape is to tamper with every control equally, which a limited reach doesn’t allow.



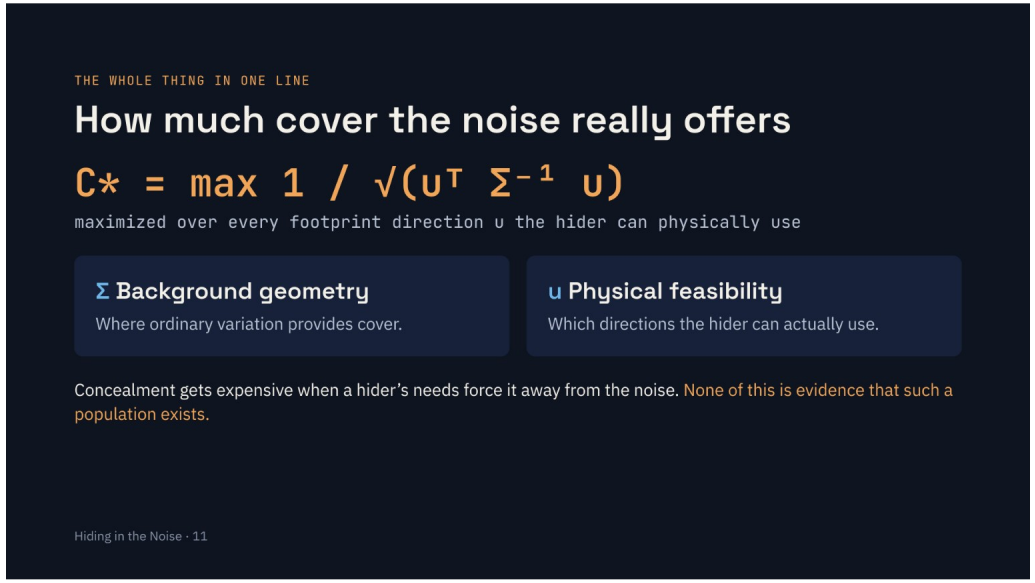
Every test has a tradeoff. One kind makes tampering worthless but is weaker against plain activity; another is the reverse. A single fixed test forces you to pick a point on that tradeoff. If you randomly choose which test runs, the hider has to commit to one strategy without knowing which test it will face, and it cannot tailor its tampering. In the model that cut the hider’s room by about 28 percent at the same level of tamper-proofing. It is the same lesson twice: randomize which comparison groups you use, and which tests you run.



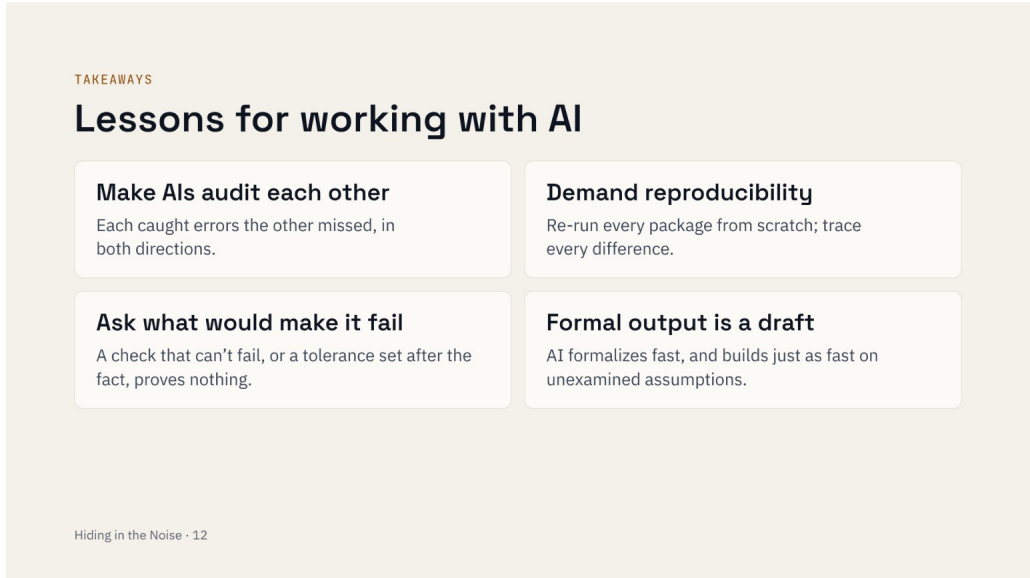
When we let the hider choose freely between tampering, hiding in noise and simply doing less, doing less usually won against a capable observer. Tampering is attractive only when it is cheap, and cross-checked tests leave it so little value that even small costs deter it. One honest caveat from our own retractions: when tampering is free, the hider uses it, and how much it does less depends on what it gains from activity. So the practical message is about suppression: good detection shrinks bad activity more often than it catches the actor.



Applied back to the aliens, the framework says: don't look at people. Under the premises they are ordinary by design, so non-detection there is exactly what you'd expect. Shared infrastructure is fixed in place, can't be randomized, and accumulates evidence over time, so persistent input-output mismatches are the most tractable target. The third signal is behavioral: anything that adapts to being watched will move when you vary observation unpredictably. Noise doesn't care whether you're looking. And the base-rate line is the ethical point: with a thousand hypothetical residents among eight billion people, even a one-in-ten-thousand false-alarm rate flags 800,000 ordinary humans, mostly the already-marginalized. Detection, if possible at all, lives in aggregates, never in a list of suspects.



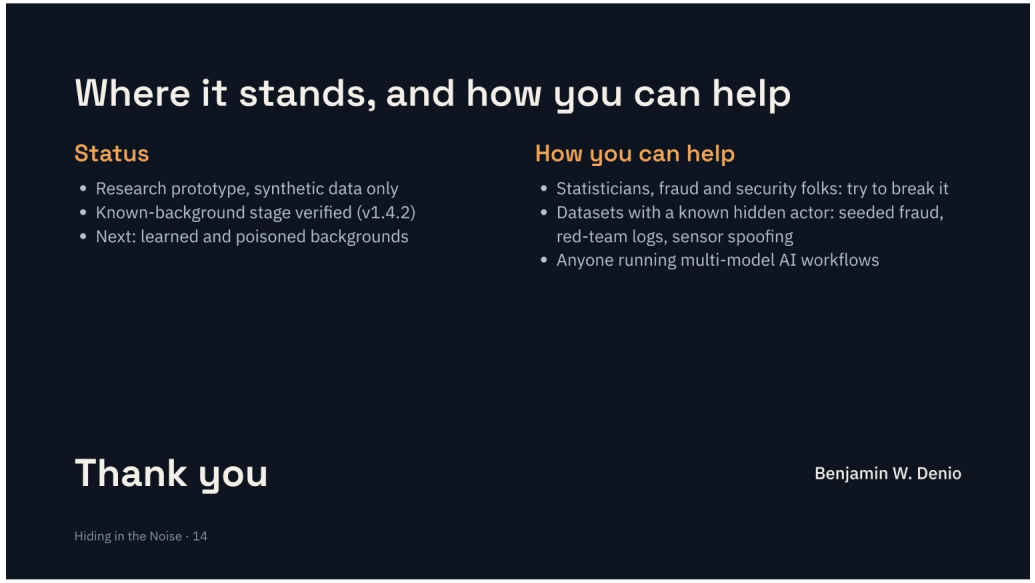
If you want one formula to take home, it's this. Sigma describes how ordinary activity varies and co-varies across all the channels you measure. The vector u is the direction a hider's activity has to push those channels. Cover is large when ordinary noise is large in exactly that direction, after everything you've calibrated. So a busy city with well-modeled, tightly correlated noise can offer less cover than a quiet place with one poorly understood channel. The hider's physical needs restrict which directions it can use; that's what makes hiding costly. And the caveat stays on the slide: this describes what evidence would have to look like, not evidence that anything is there.



These are the habits that made the difference, and they apply to any serious AI-assisted work. Two models checking each other caught things neither caught alone, and both were wrong at some point. Reproducibility meant every package was re-run from a clean folder, and one non-reproducible file turned out to be a real bug in how random numbers were drawn. The most valuable question was always “what would make this fail?” And the biggest risk: AI turns a vague idea into confident-looking math very quickly, and will build on an unchecked assumption just as quickly.



None of this is new in isolation, and that’s the reassuring part. Huber gave us the contamination model in 1964; Cachin framed hiding as hypothesis testing; Rousseeuw and Van Driessen gave us robust covariance; Axelsson showed why rare-event detection drowns in false alarms. Ecologists know that not detecting something isn’t the same as its absence. Dalvi and colleagues made the adversary a player in the game, Kloft and Laskov did it for anomaly detectors specifically, and Li and Yu bring it up to date. The full annotated list of thirty, across adversarial ML, robust statistics, sensor fusion and even camouflage biology, is in the repo.



Close with status and the ask. It’s a research prototype on synthetic data; the stage where the observer knows the background exactly is verified, and the next stage asks what happens when the background must be learned, and when a hider can fake it during learning. The most useful help: people who will try to break it, and datasets where the hidden actor is known, such as seeded fraud records, red-team logs or sensor-spoofing trials, so we can test whether the framework’s predictions hold up where the answer is known.